

Astronomy 102 - Lab 02

- Reason for the Seasons



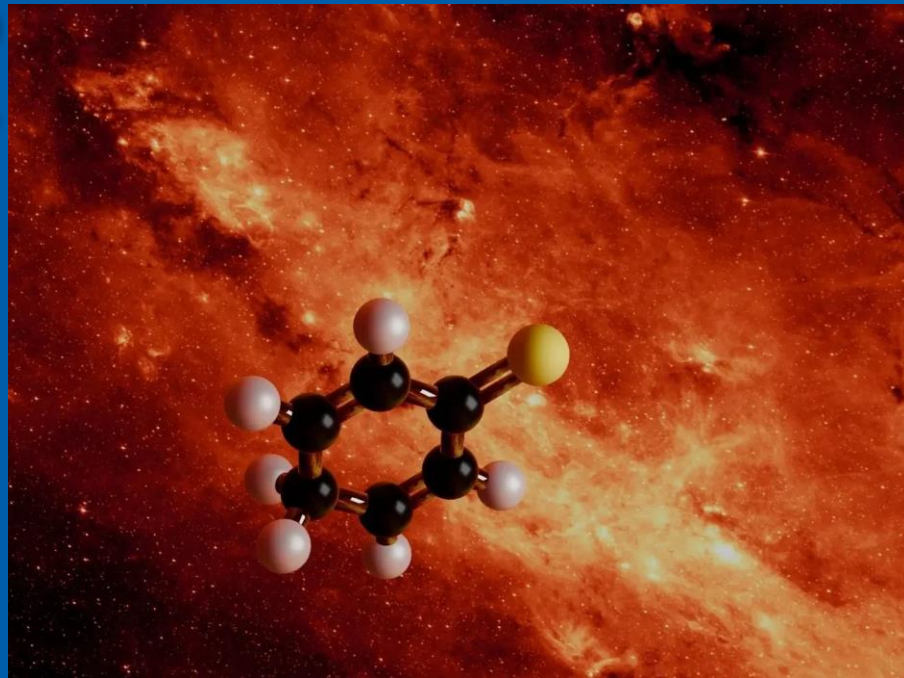
Central Park from above

The latest in Astronomy

For the First Time, Scientists Detect Molecule Critical to Life in Interstellar Space:

- Thiepine, or 2,5-cyclohexadiene-1-thione (C_6H_6S), a ring-shaped sulfur-bearing hydrocarbon produced in biochemical reactions
- A building block for cells

How do you think we can detect things as small as a molecule from outside our own solar system?



Syllabus Review

Instructor: Matthew Prest

Section: 04 (5763)

Time: Thursdays 12:00pm - 2:50pm

Office: North Building 1220A (<https://us06web.zoom.us/j/3163865896>)

Email: mprest@gradcenter.cuny.edu

Virtual Office Hours: Mondays 4:00pm - 6:30pm (<https://us06web.zoom.us/j/3163865896>)

In-person Office Hours: Appointments welcome, send me an email!

Department Office Information: Room 1225 HN, Tel: (212) 772-5248.

In-Person Lab Location: HN-C115

Reminder: No lab next week (Feb 12, HW still due!), the following week (Feb 19) will be our first in-person class!

Syllabus Review

1. **Sign up for the Pearson online textbook!** (instructions in accompanying document, and attempt lab 00 for extra credit!)
2. **Print In-person lab documents** (available on Brightspace)
3. **Add the following in-person lab dates to your calendar:**
 - a. **Thursday Feb 19th** (Lab 3, Diameter of the Moon)
 - b. **Thursday March 5th** (Lab 5, Simple Pendulum)
 - c. **Thursday March 26th** (Lab 8, Spectroscopy)
 - d. **Thursday April 23rd** (Lab 10, Lenses)
 - e. **Thursday May 14th** (Make-up Lab Session)

Lab Reports

What makes a good lab report?

- Precise instructions (what did you actually do?)
 - Don't worry if you got the "wrong" result!
- Impersonal descriptions
 - It's great that you had fun doing the experiment but we don't need to record that in the lab report! Focus on the details!
 - Proper grammar, spelling and punctuation!

▼ Part A - Aim

In your own words, describe the aim of the activity. What are you expecting to learn? This is described in the [Introduction \(p.1\)](#) and [Objectives \(p.3\)](#). Use first-person "I" statements to set an intention and take ownership of your learning.

► [View Available Hint\(s\)](#)

Essay answers are limited to about 500 words (3800 characters maximum, including spaces).

3800 Character(s) remaining

Submit

Lab Reports

A good lab report is like a good recipe!

- Easy to follow, clear instructions with all of the details you need to recreate the dish exactly as you made it, even if it didn't turn out perfect!
- That's how scientists build trust with one another, by doing experiments that are reproducible!



Pearson Online Platform

Launch the [Seasons](#) tutorial. As you go through the tutorial, answer the ungraded questions in the tutorial and the graded essay questions below.

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Astronomy Self-Guided Tutorial

Seasons

Help Page List Credits

Introduction

Objectives

Lesson 1: Factors Affecting Seasonal Changes

Lesson 2: The Solstices and Equinoxes

Lesson 3: The Sun's Position in the Sky

Revisit Prediction

Summary





Pearson Online Platform

Introduction

Objectives

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Reason for the Seasons

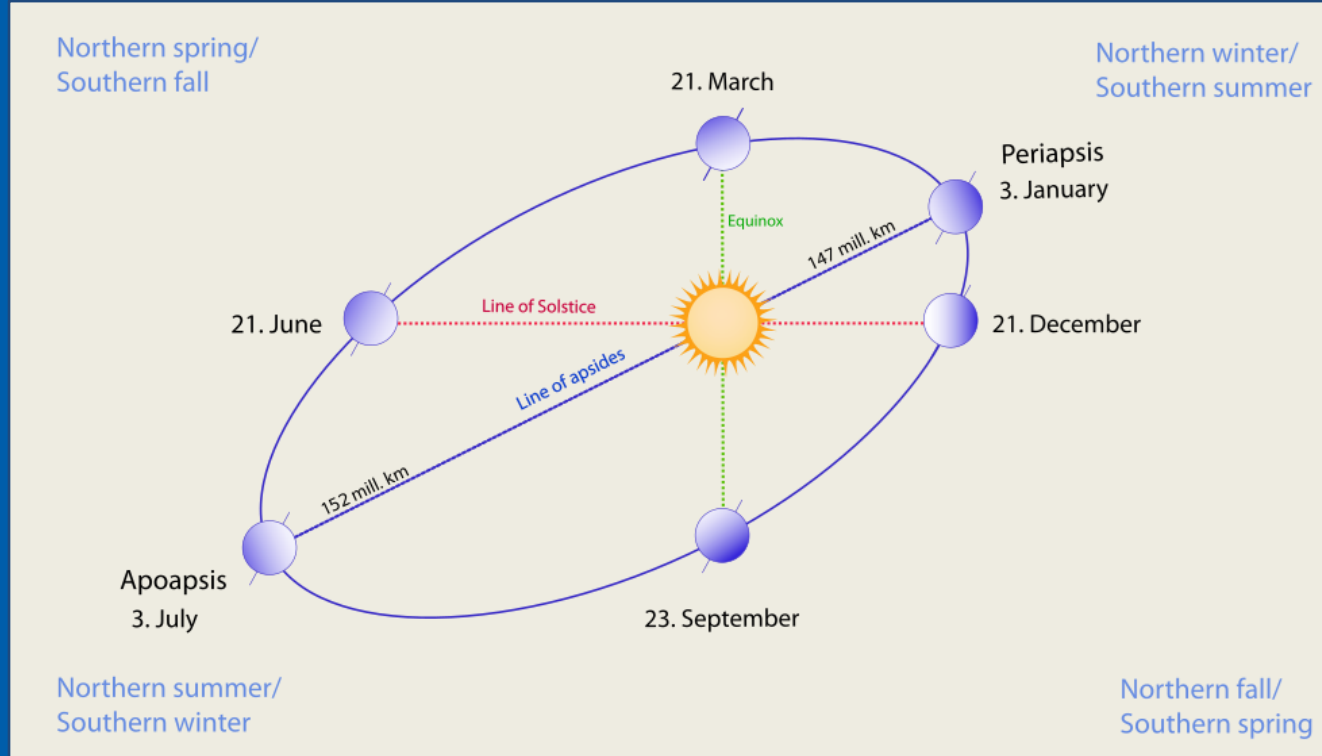
What we will cover:

- Elliptical orbits
- Directness of sunlight
- Axial tilt
- Solstice & Equinox

Reason for the Seasons

Elliptical orbits:

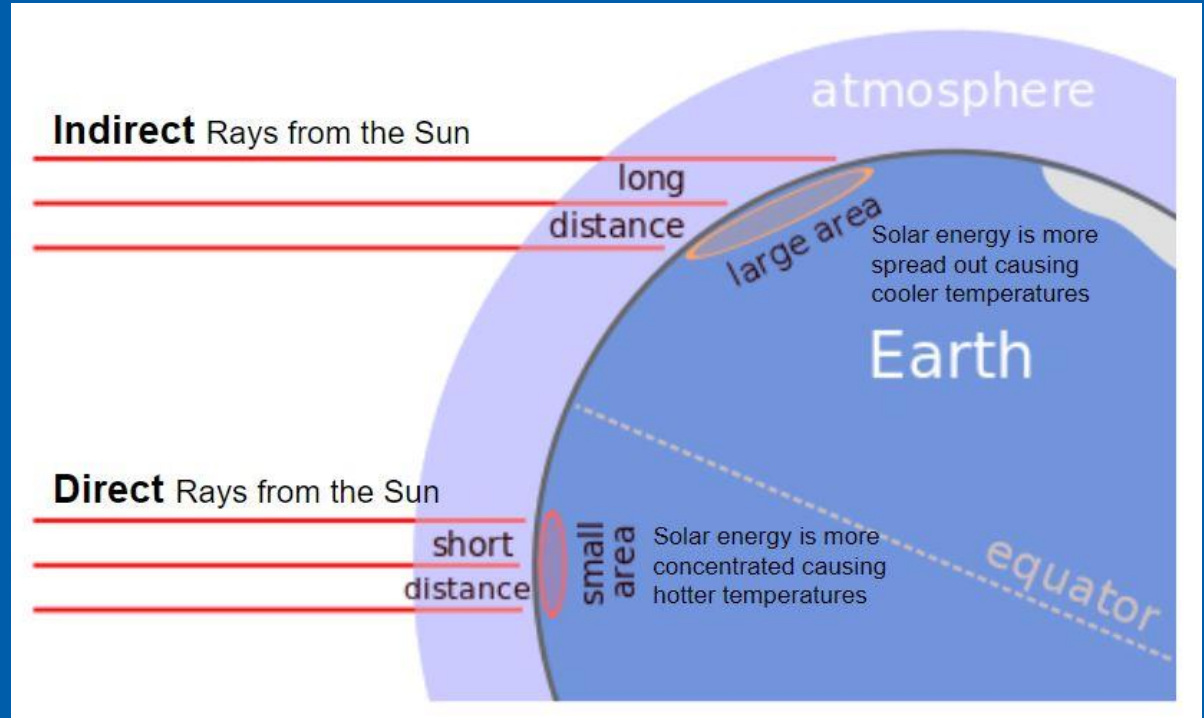
- Planets, moons, comets and all objects orbit in approximate ellipses
- The distance to the Sun can change by 3 million miles!



Reason for the Seasons

Directness of sunlight:

- Our atmosphere is really good at scattering light away
- The shorter the travel distance through our atmosphere the more intense the sunlight



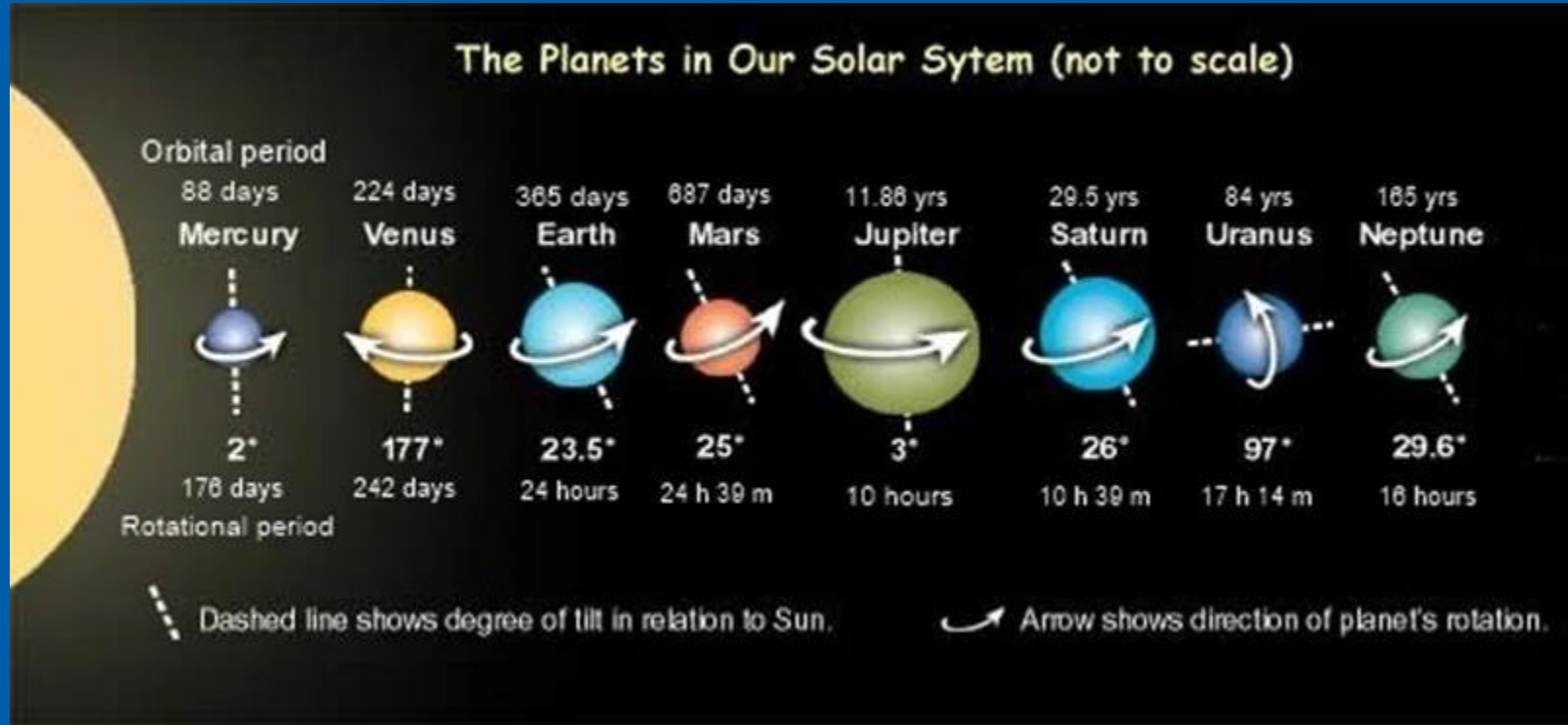
Reason for the Seasons

Axial tilt:

- Earth spins around its pole while it orbits the Sun
- Relative angle of 23.5 degrees
- This changes which hemisphere is tilted towards the Sun throughout the year



Reason for the Seasons

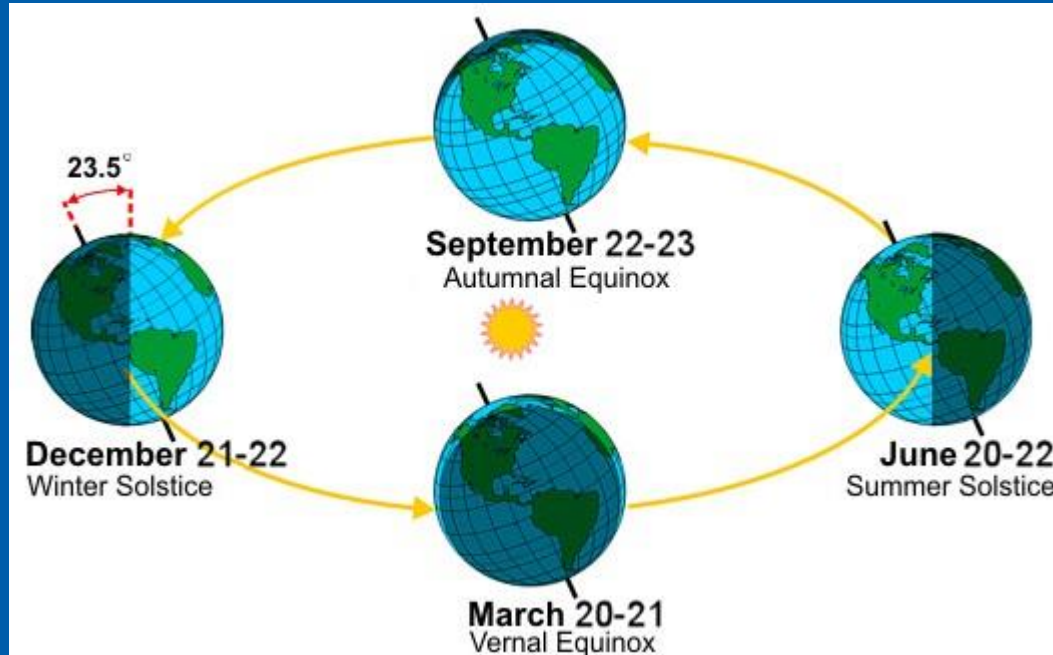


What do you notice when you compare the planets?

Reason for the Seasons

Solstice and Equinox:

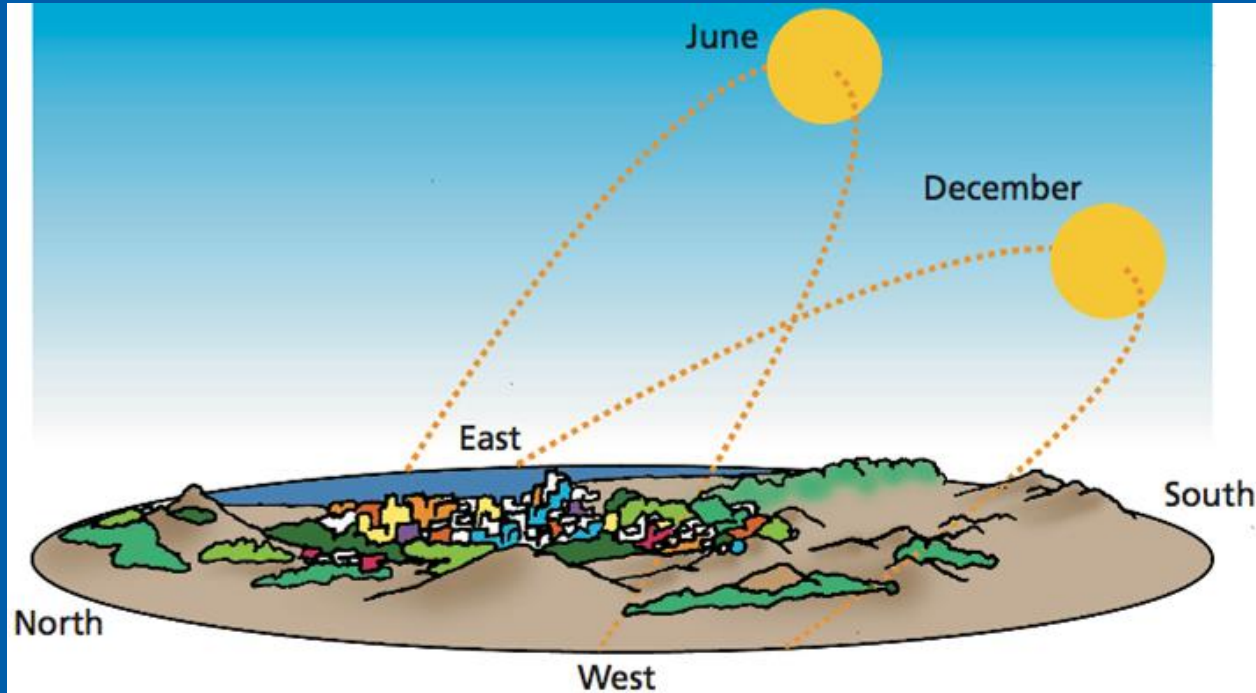
- Solstice: Shortest/Longest Day of the year, Sun reaches Highest/Lowest point
- Equinox: Equal length Day & Night, Sun directly above the equator



Reason for the Seasons

Solstice and Equinox:

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Reminders

1. **Complete the Tutorial first!**
 - This is best done now in class where I am available to assist
1. **Attempt the essay questions last!**
 - These should be based on what you have covered in the tutorial.
1. **Previous week's lab report due at the beginning of the next lab!**
 - Final grades are based on best 6 online labs plus each of the 4 in-class labs
1. **Grades will be posted to Brightspace**
 - Check now that you have access to our course!
 - Grade feedback will also be provided on Pearson, please check as this is the best way to improve your future grades!
1. **Office hours**
 - Office hours are available on Zoom Mondays 4-6pm (same zoom link as class) as well as by appointment either in-person on or zoom. Come say hi, I'm here to help!

Any and all questions are welcome!